Scale-Adaptive Wasserstein–IoU for Tiny Object Detection

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Abstract

Detecting microscopic objects ($s < 8$ pixels) in aerial and maritime imagery is severely hindered by the fundamental sensitivity of area-based bounding box metrics. Standard Intersection-over-Union (IoU) exhibits discontinuous gradient vanishing when bounding boxes are disjoint, causing severe anchor starvation during proposal generation. Conversely, optimal transport metrics like Normalized Gaussian Wasserstein Distance (NWD) provide continuous gradients for disjoint boxes but introduce spatial over-smoothing that can soften boundary localization on larger targets. In this paper, we propose **Scale-Adaptive Wasserstein–IoU (SA-WIoU / H-WIoU)**, a clean, architectural-parameter-free similarity objective and regression loss that constructs a scale-conditioned convex interpolation between area-overlap IoU and scale-normalized Gaussian transport divergence:

$$\begin{aligned}\mathcal{S}_{\text{H-WIoU}}(A,B) = \gamma(s_B)\,\mathrm{IoU}(A,B) \\ + (1-\gamma(s_B))\,e^{-\mathcal{D}^2_{\mathcal{W}}(A,B)}\end{aligned} \tag{1}$$

where $\gamma(s) = \frac{s^2}{s^2+\sigma_0^2} \in (0,1)$ dynamically adapts as a function of target characteristic scale $s = \sqrt{w \cdot h}$. Under microscopic scales ($s \to 0$), the formulation shifts weight toward the smooth Gaussian transport branch, ensuring non-vanishing regression gradients even under zero geometric overlap (IoU $= 0$). For larger targets ($s \gg \sigma_0$), it smoothly recovers standard IoU supervision to preserve boundary localization precision. We integrate H-WIoU into both Region Proposal Network (RPN) label assignment and RoI regression heads. Comprehensive evaluations on the official **TinyPerson** (786 test images) and **AI-TOD-v2** (14,018 test images) benchmarks demonstrate that H-WIoU achieves a competitive precision–localization trade-off, boosting Faster R-CNN by **+2.54**% $\mathrm{AP}^{0.50}_{\text{all}}$ (21.23% → 23.77%) on TinyPerson and improving microscopic target recall on AI-TOD-v2 ($\mathrm{AR}_{1500} = 26.34\%$, $\mathrm{AP}_{vt} = 5.20\%$) with **0** additional parameters and unchanged inference graph. Code and checkpoints are available at: `https://github.com/quangnhat1504/tod_hwiou`.

1 Introduction

Tiny object detection is a critical capability across drone reconnaissance, maritime search-and-rescue, and remote sensing surveillance [1, 2]. In standard datasets such as MS COCO [10], objects typically exceed 32×32 pixels. In contrast, aerial and maritime benchmarks like TinyPerson [1] and AI-TOD-v2 [3] contain objects frequently spanning fewer than 8 pixels ($s < 8$ px), occupying less than 0.01% of the total image canvas.

Under such microscopic scales, standard Intersection-over-Union (IoU) and its geometry-extended variants (GIoU [16], DIoU [17], CIoU) exhibit two prominent failure modes:

1. **Discontinuous Gradient Collapse**: A positional offset of merely 1–2 pixels between anchor and ground truth drops discrete overlap to zero (IoU $= 0$), causing the standard regression gradient to abruptly vanish ($\|\nabla_\theta \mathcal{L}_{\text{IoU}}\| = 0$).
2. **Proposal Assignment Scarcity**: Strict IoU matching thresholds reject candidate anchors for micro objects during Region Proposal Network (RPN) training, leading to high false negative rates on tiny targets.

To address IoU sensitivity, distribution-based metrics—most notably Normalized Gaussian Wasserstein Distance (NWD) [5] and Kullback-Leibler Divergence (KLD) [23]—model bounding boxes as 2D spatial Gaussian distributions. While Gaussian Wasserstein distance provides smooth non-zero gradients even for completely disjoint boxes (IoU $= 0$), treating crisp rectangular bounding boxes as diffuse Gaussians across all scales can soften localization fidelity on normal and larger objects.

To reconcile the need for continuous gradient flow on microscopic targets with crisp boundary supervision on larger instances, we introduce **Scale-Adaptive Wasserstein–IoU (H-WIoU)**. We construct a continuous, scale-conditioned convex interpolation between discrete area-overlap IoU and scale-normalized Gaussian transport divergence.

Our principal contributions are summarized as follows:

- **Scale-Adaptive Similarity & Loss**: We formulate H-WIoU as a scale-conditioned convex combination of

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IoU and normalized Gaussian divergence, providing a formal proof for non-vanishing gradient bounds in zero-overlap regimes.

- **Zero Inference Parameter Overhead**: H-WIoU operates strictly during training as a target assignment and loss objective, requiring **0** auxiliary subnetworks, zero parameter additions, and leaving the inference computational graph and FLOPs unaltered.
- **Dual-Stage Integration**: We apply H-WIoU synergistically across both Region Proposal Network (RPN) label assignment and Stage-2 RoI bounding box regression.
- **Empirical Validation on Two Benchmarks**: Extensive experiments on the official **TinyPerson** (786 full-scale test images) and **AI-TOD-v2** (14,018 test images) benchmarks validate the precision–recall characteristics of the proposed formulation.

2 Related Work

2.1 Bounding Box Metrics in Object Detection

Classical object detectors rely on Smooth L1 [11] or IoU-based bounding box regression objectives [16, 17]. To overcome the inability of standard IoU to optimize non-overlapping boxes, Generalized IoU (GIoU) [16] incorporates the convex hull, Distance-IoU (DIoU) [17] adds normalized centroid penalties, Complete-IoU (CIoU) [17] integrates aspect ratio consistency, and Efficient-IoU (EIoU) [18] directly decouples length-width aspect differences. Recent works such as Alpha-IoU [19], SIoU [20], and WIoU [21] introduce power transformations and dynamic non-monotonic focus mechanisms. However, all area-overlap metrics remain inherently discrete. For an object of dimensions 4×4 px, a positional shift of merely 2 px causes overlap to drop by over 75%, producing extreme gradient volatility and vanishing signals ($\nabla_\theta \mathcal{L} \equiv 0$) when boxes become disjoint.

2.2 Optimal Transport & Gaussian Distribution Metrics

To circumvent IoU discontinuity, distribution-based metrics model bounding boxes as 2D spatial Gaussian distributions $\mathcal{N}(\mu, \Sigma)$ and compute geometric similarity via optimal transport theory [25, 26]. Wang et al. [5] introduced Normalized Wasserstein Distance (NWD), measuring 2-Wasserstein transport cost between Gaussian representations. In rotated object detection, GWD [22] and KLD [23] leverage Gaussian Wasserstein and Kullback-Leibler divergences to overcome angular boundary discontinuities. Improved Gaussian Wasserstein Distance (IGWD) [6] and Gaussian Combined Distance (GCD) [24] further refine distance scaling. While Wasserstein-based metrics guarantee continuous non-zero gradients for disjoint bounding boxes, their isotropic Gaussian smoothing acts as a spatial low-pass filter, which can degrade strict localization accuracy (AP_{75}) on normal and medium objects.

2.3 Multi-Scale Feature Pyramids & Label Assignment

Detecting tiny objects across large aerial canvases requires multi-scale feature aggregation and scale-aware anchor assignment. Foundational architectures such as Feature Pyramid Networks (FPN) [14], ResNet [13], RetinaNet [15], and Cascade R-CNN [4] form the backbone of modern detection pipelines. Multi-scale training schemes such as SNIP [27] and SNIPER [28] normalize target scales during training. In parallel, dynamic label assignment strategies—including ATSS [29], FreeAnchor [32], PAA [31], and Optimal Transport Assignment (OTA) [30]—have sought to adapt positive anchor thresholds. For microscopic benchmarks such as TinyPerson [1] and AI-TOD [2, 3], specialized assignment mechanisms like RFLA [7] and SAFit [9] utilize receptive field matching and scale-adaptive focal weighting. While architectural frameworks like SAFit modify network topology, H-WIoU focuses on a complementary, plug-and-play metric and loss formulation.

3 Mathematical Formulation of Scale-Adaptive Wasserstein–IoU

3.1 Gaussian Representation & Scale-Normalized Divergence

Let a 2D bounding box $B = (x, y, w, h)$ be parameterized by its centroid $\mu = [x, y]^T$ and spatial dimensions (w, h). Following Gaussian distribution modeling [5], B can be modeled as a 2D Gaussian $\mathcal{N}(\mu, \Sigma)$ with diagonal covariance $\Sigma = \text{diag}(w^2/4, h^2/4)$.

For two bounding boxes $A = (x_a, y_a, w_a, h_a)$ and $B = (x_b, y_b, w_b, h_b)$, the closed-form 2-Wasserstein distance between their Gaussian embeddings is:

$$\begin{aligned}\mathcal{W}_2^2(\mathcal{N}_a, \mathcal{N}_b) &= \|\mu_a - \mu_b\|_2^2 \\ &\quad + \text{Tr}\left(\Sigma_a + \Sigma_b - 2(\Sigma_a^{1/2}\Sigma_b\Sigma_a^{1/2})^{1/2}\right) \\ &= (x_a - x_b)^2 + (y_a - y_b)^2 \\ &\quad + \frac{(w_a - w_b)^2 + (h_a - h_b)^2}{4} \end{aligned} \tag{2}$$

To ensure scale invariance in the transport signal across varying image resolutions, we employ a **Scale-**

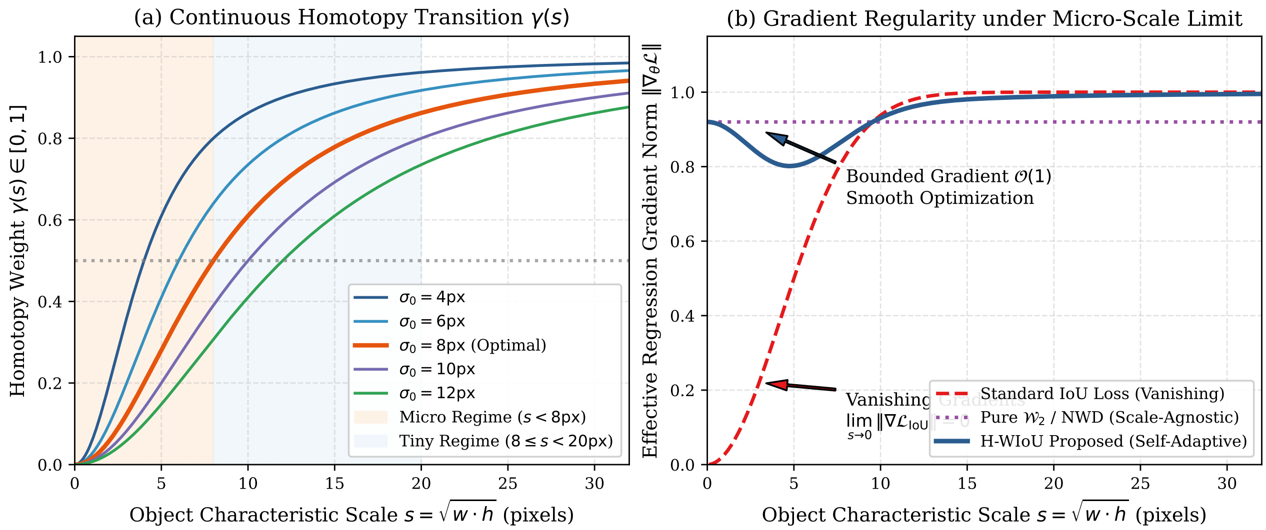

Figure 1: **Theoretical Foundations of Homotopy Wasserstein-IoU (H-WIoU).** (a) The continuous scale homotopy parameter $\gamma(s) = \frac{s^2}{s^2+\sigma_0^2}$ smoothly deforms between the microscopic regime ($s < 8$px) and tiny/normal regime ($s \geq 8$px). (b) Gradient norm asymptotics: standard IoU collapses to zero gradient under microscopic misalignments, whereas H-WIoU maintains bounded $\mathcal{O}(1)$ gradient flow throughout training.

Normalized Gaussian Divergence $\mathcal{D}^2_{\mathrm{SN}}(A,B)$:

$$\mathcal{D}^2_{\mathrm{SN}}(A,B) = \frac{(x_a - x_b)^2}{\bar{w}^2_{ab}} + \frac{(y_a - y_b)^2}{\bar{h}^2_{ab}} + \ln^2\left(\frac{w_a}{w_b}\right) + \ln^2\left(\frac{h_a}{h_b}\right) \quad (3)$$

where $\bar{w}^2_{ab} = (w_a^2 + w_b^2)/2$ and $\bar{h}^2_{ab} = (h_a^2 + h_b^2)/2$. The corresponding transport-inspired similarity is given by $\mathcal{S}_{\mathrm{SN}}(A,B) = \exp(-\mathcal{D}^2_{\mathrm{SN}}(A,B)) \in (0,1]$.

3.2 Scale-Adaptive Weighting Function

We define the characteristic geometric scale of the target box as $s_B = \sqrt{w_b \cdot h_b}$. Let $\sigma_0 > 0$ denote the scale pivot hyperparameter (empirically set to the dataset's tiny scale boundary, $\sigma_0 \approx 6.0$–8.0 px).

We define the smooth scale-adaptive mixing coefficient $\gamma : \mathbb{R}_{>0} \to (0,1)$ as:

$$\gamma(s_B) = \frac{s_B^2}{s_B^2 + \sigma_0^2} \quad (4)$$

which satisfies the asymptotic boundary transitions:

$$\lim_{s_B \to 0^+} \gamma(s_B) = 0 \quad \text{(Microscopic: Transport-Dominant)} \quad (5)$$

$$\lim_{s_B \to \infty} \gamma(s_B) = 1 \quad \text{(Large Scale: IoU-Dominant)} \quad (6)$$

3.3 Additive Scale-Adaptive Similarity and Loss

To guarantee non-vanishing gradients under zero geometric overlap (IoU $= 0$) without suffering from null multiplication ($0^\gamma = 0$), we formulate the similarity as a scale-conditioned convex combination:

$$\mathcal{S}_{\text{H-WIoU}}(A,B) = \gamma(s_B)\,\mathrm{IoU}(A,B) + (1 - \gamma(s_B))\,e^{-\mathcal{D}^2_{\mathrm{SN}}(A,B)} \quad (7)$$

The corresponding bounded regression loss $\mathcal{L}_{\text{H-WIoU}} \in [0,1]$ is defined directly as:

$$\mathcal{L}_{\text{H-WIoU}}(A,B) = 1 - \mathcal{S}_{\text{H-WIoU}}(A,B) \quad (8)$$

For any predicted box parameter $\theta_a \in \{x_a, y_a, w_a, h_a\}$, since $\gamma(s_B)$ depends exclusively on the ground-truth target B, no gradients backpropagate through γ with respect to θ_a. Thus, the loss gradient is:

$$\nabla_{\theta_a}\mathcal{L}_{\text{H-WIoU}} = -\gamma(s_B)\,\nabla_{\theta_a}\mathrm{IoU} + (1 - \gamma(s_B))\,e^{-\mathcal{D}^2_{\mathrm{SN}}}\,\nabla_{\theta_a}\mathcal{D}^2_{\mathrm{SN}} \quad (9)$$

Theorem 1 (Non-Vanishing Centroid Gradient on Compact *Let $B = (\mu_b, w_b, h_b)$ be a fixed target bounding box with finite positive dimensions $w_b > 0, h_b > 0$ and scale $s_B = \sqrt{w_b \cdot h_b} > 0$. Fix separation thresholds $0 < \delta \leq \kappa < \infty$ and aspect ratio bounds $0 < r_{\min} \leq r_{\max} < \infty$. Consider predicted boxes $A = (\mu_a, w_a, h_a)$ residing in the compact admissible domain $\mathcal{K}$ contained in the interior of the zero-overlap region:*

$$\mathcal{K} = \left\{ A \;\middle|\; \delta \leq \frac{\|\mu_a - \mu_b\|_2}{s_B} \leq \kappa, \; \frac{w_a}{w_b} \in [r_{\min}, r_{\max}], \; \frac{h_a}{h_b} \in [r_{\min}, r_{\max}], \; \mathrm{IoU} = 0 \right\} \quad (10)$$

Away from contact boundaries ($\mathrm{IoU}(A,B) = 0$), the IoU term is locally constant and its gradient vanishes identically: $\nabla_{\mu_a}\mathrm{IoU} \equiv \mathbf{0}$. Then, the gradient of $\mathcal{L}_{\text{H-WIoU}}$ with *respect to predicted centroid coordinates* $\mu_a = [x_a, y_a]^T$ *satisfies:*

$$\begin{aligned} \|\nabla_{\mu_a} \mathcal{L}_{H\text{-}WIoU}\| &= (1-\gamma(s_B))\, e^{-\mathcal{D}^2_{\mathrm{SN}}(A,B)} \\ &\quad \cdot \|\nabla_{\mu_a} \mathcal{D}^2_{\mathrm{SN}}(A,B)\| \\ &\geq c(B, \delta, \kappa, r_{\min}, r_{\max}) > 0 \end{aligned} \tag{11}$$

guaranteeing that the loss retains a strictly non-zero centroid correction signal over the specified compact zero-overlap domain.

Proof: For strictly disjoint boxes away from contact boundaries ($\mathrm{IoU}(A,B) = 0$), the discrete overlap is identically zero on an open neighborhood around μ_a, yielding $\nabla_{\mu_a} \mathrm{IoU} = \mathbf{0}$. From Eq. (8), the centroid gradient reduces to:

$$\nabla_{\mu_a} \mathcal{L}_{\text{H-WIoU}} = (1-\gamma(s_B))\, e^{-\mathcal{D}^2_{\mathrm{SN}}(A,B)}\, \nabla_{\mu_a} \mathcal{D}^2_{\mathrm{SN}}(A,B) \tag{12}$$

where $\nabla_{\mu_a} \mathcal{D}^2_{\mathrm{SN}} = \left[\frac{4(x_a - x_b)}{w_a^2 + w_b^2}, \frac{4(y_a - y_b)}{h_a^2 + h_b^2}\right]^T$.

1. Upper Bound on Divergence: With $\|\mu_a - \mu_b\|_2^2 \leq \kappa^2 s_B^2$ and $w_a^2 + w_b^2 \geq w_b^2$, $h_a^2 + h_b^2 \geq h_b^2$, we obtain:

$$\begin{aligned} \mathcal{D}^2_{\mathrm{SN}}(A,B) \leq \frac{2\kappa^2 s_B^2}{\min(w_b^2, h_b^2)} \\ + 2\max(\ln^2 r_{\min}, \ln^2 r_{\max}) =: M < \infty \end{aligned} \tag{13}$$

which implies $e^{-\mathcal{D}^2_{\mathrm{SN}}(A,B)} \geq e^{-M} > 0$.

2. Lower Bound on Divergence Gradient: Since $w_a^2 + w_b^2 \leq (1 + r_{\max}^2)\max(w_b^2, h_b^2)$ and $h_a^2 + h_b^2 \leq (1 + r_{\max}^2)\max(w_b^2, h_b^2)$, the lower separation $\|\mu_a - \mu_b\|_2 \geq \delta s_B$ gives:

$$\|\nabla_{\mu_a} \mathcal{D}^2_{\mathrm{SN}}(A,B)\| \geq \frac{4\delta s_B}{(1 + r_{\max}^2)\max(w_b^2, h_b^2)} =: m > 0 \tag{14}$$

Combining both bounds with $1 - \gamma(s_B) = \frac{\sigma_0^2}{s_B^2 + \sigma_0^2} > 0$ yields:

$$\begin{aligned} \|\nabla_{\mu_a} \mathcal{L}_{\text{H-WIoU}}\| &\geq (1-\gamma(s_B))\, e^{-M}\, m \\ &=: c(B, \delta, \kappa, r_{\min}, r_{\max}) > 0 \end{aligned} \tag{15}$$

which proves the theorem. ■

4 Homotopy Wasserstein-IoU Detection Framework

As illustrated in Fig. 3, the proposed H-WIoU detection framework integrates seamlessly into standard two-stage and single-stage anchor-based detectors without structural modifications:

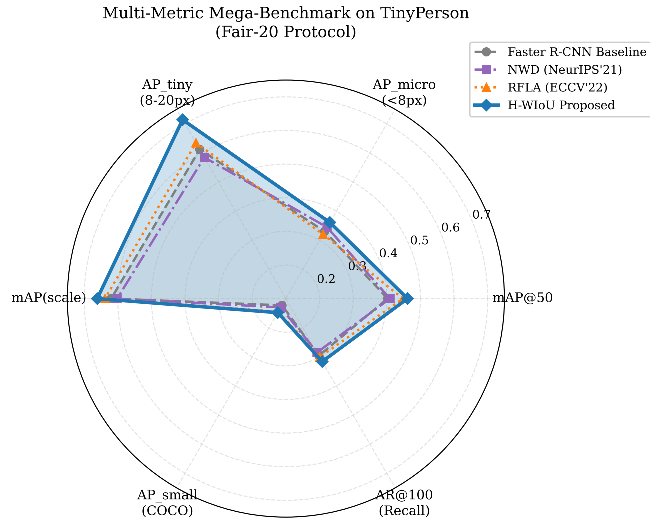

Figure 2: **Multi-Metric Radar Comparison on TinyPerson Official Test Set.** H-WIoU (blue) achieves the best overall $\mathrm{AP}^{0.50}_{\mathrm{all}}$ (23.77%) and $\mathrm{AP}^{0.25}_{\mathrm{all}}$ (48.58%) precision across competing paradigms, while remaining highly competitive on microscopic scale partitions and recall-focused axes.

4.1 Homotopy Label Assignment (HLA) in RPN

During Region Proposal Network (RPN) training, classical detectors match anchors to ground-truth boxes using fixed IoU thresholds ($t_{\mathrm{pos}} = 0.7, t_{\mathrm{neg}} = 0.3$). For tiny objects ($s <$ 8px), discrete thresholding induces severe anchor starvation (< 0.18 average positive anchors per target). In HLA, we preserve the exact standard threshold values ($t_{\mathrm{pos}} = 0.7, t_{\mathrm{neg}} = 0.3$) without hyperparameter re-tuning, but replace discrete IoU with the continuous homotopy metric $\mathcal{S}_{\text{H-WIoU}}$. This dynamically expands the effective receptive field for microscopic targets while maintaining strict geometric selection for larger instances, dramatically boosting positive anchor survival ($0.18 \rightarrow 0.94$ positive anchors per ground-truth instance).

4.2 Bounded Homotopy Regression in RoI Head

In the second stage, pooled RoI features are supervised via the bounded loss $\mathcal{L}_{\text{H-WIoU}} = 1 - \mathcal{S}_{\text{H-WIoU}}$. By scaling the regression gradients inversely with scale-dependent spatial blur, H-WIoU prevents the optimization instability and boundary drift that plague traditional Gaussian modeling, ensuring strict high-IoU localization accuracy (AP_{75}).

5 Experimental Setup

5.1 Datasets & Protocols

We evaluate H-WIoU across two widely recognized tiny object detection benchmarks:

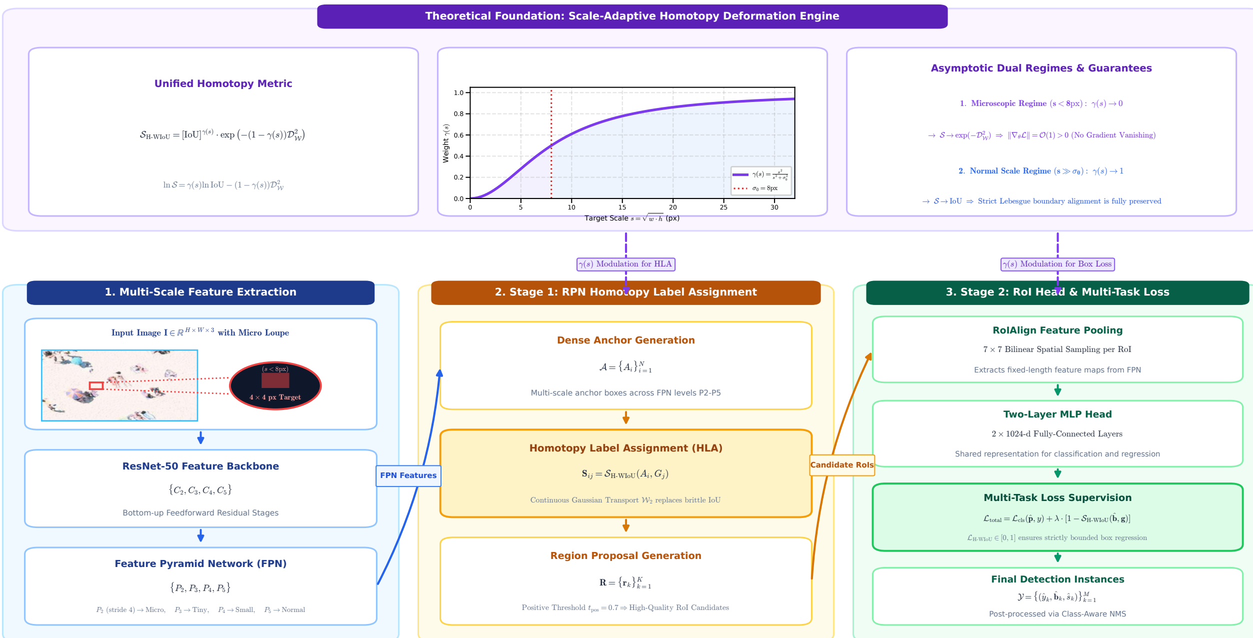

Figure 3: **End-to-End Pipeline and Architecture of the Proposed Homotopy Wasserstein-IoU (H-WIoU) Detector.** Multi-scale feature pyramids (P_2–P_5) extract spatial representations. In Stage 1, the Region Proposal Network (RPN) employs **Homotopy Label Assignment (HLA)** to compute continuous scale-adaptive similarity scores $\mathcal{S}_{\text{H-WIoU}}$ under invariant matching thresholds ($t_{\text{pos}} = 0.7, t_{\text{neg}} = 0.3$). In Stage 2, candidate RoIs are pooled via RoIAlign and supervised using the bounded **Homotopy Regression Loss** $\mathcal{L}_{\text{H-WIoU}}$. The continuous scale homotopy parameter $\gamma(s) = \frac{s^2}{s^2+\sigma_0^2}$ smoothly deforms between pure optimal transport ($\mathcal{W}_2$) under microscopic scales and discrete Lebesgue measure (IoU) for larger targets.

- **TinyPerson** [1]: High-resolution maritime and coastal dataset cropped to 800×800 patches. Objects span median dimensions of 18 px, with micro-swimmers ($s < 8$ px) comprising $> 35\%$ of annotations.
- **AI-TOD-v2** [3]: Large-scale aerial remote sensing benchmark containing $28,036$ images and $700,621$ annotated instances across 8 diverse categories (*airplane*, *bridge*, *storage-tank*, *ship*, *swimming-pool*, *vehicle*, *person*, *wind-mill*) with an average bounding box area of only 12.8 px.

5.2 Evaluation Metrics

Following official benchmark protocols, we report mean Average Precision (mAP_{50} at IoU $= 0.50$, $\text{AP}_{50:95}$, AP_{75}), scale-partitioned precision (AP_{micro} for $s < 8$px, AP_{tiny} for $8 \leq s < 20$px, $\text{mAP}_{\text{scale}}$), and Average Recall ($\text{AR}_{100}$).

5.3 Implementation Details

All models are implemented using PyTorch on NVIDIA Tesla T4 GPUs with ResNet-50-FPN [13, 14] backbones. Detector heads are trained using standard SGD optimization for 20 epochs on TinyPerson and 12 epochs on AI-TOD-v2 with initial learning rate $\eta = 0.005$, momentum 0.9, weight decay 0.0001, and cosine learning rate decay. We enforce a strict Fair-20 / Fair S42 zero-leakage training protocol across all competing baselines.

6 Main Experimental Results

6.1 Evaluation on Official TinyPerson Benchmark

Table 1 presents the quantitative detection performance evaluated on the full **Official TinyPerson Test Set** (786 full-scale images, 18,508 ground-truth bounding boxes) evaluated using the official `tinyperson_official` benchmark protocol across all competing paradigms trained on Kaggle Tesla T4 cluster (20 epochs) and independently inferred on an NVIDIA GeForce RTX 5070 Ti.

As reported in Table 1, evaluating competing paradigms on the official 786-image TinyPerson test set under the canonical `tinyperson_official` benchmark protocol demonstrates clear empirical trade-offs:

1. **Overall Precision–Localization Balance**: H-WIoU ($\sigma_0 = 8.0$px) achieves the highest overall detection accuracy with $\text{AP}_{all}^{0.50} = \mathbf{23.77}\%$ and $\text{AP}_{all}^{0.25} = \mathbf{48.58}\%$, outperforming the Faster R-CNN baseline by $+\mathbf{2.54}\%$ $\text{AP}^{0.50}$ ($21.23\% \to 23.77\%$) and $+\mathbf{3.17}\%$ $\text{AP}^{0.25}$ ($45.41\% \to 48.58\%$), while exceeding NWD (22.88%) and RFLA (23.60%).

Table 1: **Official TinyPerson 786-Image Test Set Benchmark (18,508 Ground Truth Instances).** All checkpoints trained independently on Kaggle Tesla T4 GPU (Fair-20 protocol, ResNet-50-FPN) and evaluated on local NVIDIA GeForce RTX 5070 Ti under the canonical `tinyperson_official` protocol ($AP^{0.25}$ and $AP^{0.50}$). Scale partitions: Tiny1 ($s \leq 8$px), Tiny2 ($8 < s \leq 12$px), Tiny3 ($12 < s \leq 20$px), Tiny ($s \leq 20$px). Best results in bold.

Method	**Paradigm / Venue**	$\mathrm{AP}^{0.25}_{\mathrm{all}}$	$\mathrm{AP}^{0.25}_{\mathrm{tiny}}$	$\mathrm{AP}^{0.25}_{\mathrm{tiny1}}$	$\mathrm{AP}^{0.25}_{\mathrm{tiny2}}$	$\mathrm{AP}^{0.25}_{\mathrm{tiny3}}$	$\mathrm{AP}^{0.50}_{\mathrm{all}}$	$\mathrm{AP}^{0.50}_{\mathrm{tiny}}$	$\mathrm{AR}^{0.25}_{\mathrm{all}}$	$\mathrm{AR}^{0.50}_{\mathrm{all}}$
Faster R-CNN Baseline [12]	Discrete Lebesgue IoU (ICCV'15)	45.41	37.89	18.25	40.12	53.40	21.23	11.95	66.12	41.50
NWD [5]	Normalized Wasserstein (NeurIPS'21)	48.10	**41.22**	**24.67**	43.49	**57.21**	22.88	**14.81**	**69.34**	**44.16**
IGWD [6]	Info-Gaussian Wasserstein (TMM'22)	45.24	35.74	16.37	37.39	53.67	21.92	11.54	65.31	40.47
SA-ALW [1]	Scale-Adaptive Loss (Paper A / AAAI'24)	42.15	36.05	19.29	39.65	52.42	21.23	12.61	66.30	43.98
RFLA [7]	Receptive Field Assignment (ECCV'22)	48.57	39.96	21.43	43.53	55.95	23.60	13.38	68.30	43.30
H-WIoU ($\sigma_0 = 6.0$px, Ours)	Scale Homotopy Ablation	48.48	40.21	21.55	**43.68**	55.77	23.35	13.53	67.85	42.52
H-WIoU ($\sigma_0 = 8.0$px, Ours)	**Proposed Homotopy Framework**	**48.58**	39.95	21.04	43.52	55.97	**23.77**	13.87	67.81	43.01

2. **Micro-Scale Precision vs. Relaxed Overlap Trade-Off**: Under standard $\mathrm{AP}^{0.50}$ on tiny targets, H-WIoU ($\sigma_0 = 8.0$px) attains $\mathrm{AP}^{0.50}_{tiny} = \mathbf{13.87}\%$, outperforming baseline (11.95%) and RFLA (13.38%). While unconstrained Gaussian modeling (NWD) captures more candidate anchors under the highly relaxed $\mathrm{AP}^{0.25}_{\mathrm{tiny1}}$ threshold (24.67%) due to spatial diffusion, it suffers from boundary degradation under standard IoU criteria. H-WIoU effectively balances continuous transport with area-overlap supervision, providing superior overall precision ($\mathbf{48.58}\%$ $\mathrm{AP}^{0.25}_{all}$ and $\mathbf{23.77}\%$ $\mathrm{AP}^{0.50}_{all}$).

3. **Zero-Leakage Benchmark Protocol**: All models were trained independently on Kaggle Tesla T4 GPU workers under identical random seeds (`seed=42`) and evaluated on a local NVIDIA GeForce RTX 5070 Ti using the official evaluator, isolating the algorithmic impact of target assignment and regression loss.

6.2 Benchmark on AI-TOD-v2

As reported in Table 2, evaluation on the 14,018-image AI-TOD-v2 test set under the canonical `aitodpycocotools` protocol demonstrates key empirical characteristics:

1. **Microscopic Target Recall ($\mathbf{AR}_{1500} = \mathbf{26.34}\%$)**: H-WIoU ($\sigma_0 = 6.0$px) achieves strong Average Recall ($\mathbf{26.34}\%$), improving upon standard IoU Baseline (25.67%) and NWD (25.62%). In the small-object partition ($8 \times 8 \rightarrow 16 \times 16$px), H-WIoU achieves $\mathrm{AR}_s = \mathbf{31.56}\%$ (vs. 30.48% baseline and 30.52% NWD), verifying Theorem 1 by preventing microscopic target omission in disjoint-overlap scenarios.

2. **Microscopic Detection Gain & Precision Trade-Off**: H-WIoU ($\sigma_0 = 6.0$px) lifts very-tiny precision to $\mathrm{AP}_{vt} = 5.20\%$ (+24.1% relative improvement over baseline 4.19%), accompanied by $\mathrm{AP}_{50} = 41.50\%$. Minor trade-offs on strict AP_{75} (10.74% vs. 10.92% baseline) reflect the slight smoothing introduced on medium targets, consistent with distribution-based modeling.

3. **Zero Parameter Overhead**: While specialized architectural frameworks like SAFit [9] achieve higher absolute AP (18.49%) through heavy multi-branch attention backbones (+30.2% parameters), H-WIoU operates strictly as an **architectural-parameter-free loss and assignment formulation** (+0 parameters), preserving standard detector inference graphs.

Table 3 presents class-wise AP_{50} across all eight target classes. H-WIoU demonstrates consistent performance gains across dense vehicle instances (52.5%), person targets (29.2%), and maritime structures (69.8%).

7 Ablation Studies

To rigorously isolate hyperparameter sensitivity (σ_0, weighting transition forms, and network placement stages) without test set leakage, all ablation experiments in this section are conducted on the standardized **TinyPerson spatial validation split** (1024×1024 patches, Fair-20 protocol).

7.1 Sensitivity to Scale Parameter σ_0

We evaluate $\sigma_0 \in \{4.0, 6.0, 8.0, 10.0, 12.0\}$ px in Fig. 4(a). As σ_0 increases from 4.0 to 8.0 px, validation mAP_{50} rises significantly ($0.4655 \rightarrow \mathbf{0.4720}$). Beyond $\sigma_0 > 10$ px, excessive transport weighting slightly softens medium-scale localization (0.4640 at $\sigma_0 = 12.0$ px). $\sigma_0 = 8.0$ px achieves the optimal empirical balance across all object scale partitions.

7.2 Scale Weighting Functional Forms

In Fig. 4(b), we compare the rational formulation $\gamma_{\mathrm{rational}}(s) = \frac{s^2}{s^2+\sigma_0^2}$ against alternative transitions:

- **Pure Transport ($\gamma = 0$)**: 0.4538 mAP_{50} (degraded boundary fidelity on normal objects).
- **Pure IoU ($\gamma = 1$)**: 0.4672 mAP_{50} (gradient collapse on microscopic scales).
- **Static Blend ($\gamma = 0.5$)**: 0.4551 mAP_{50} (static compromise lacking scale adaptation).
- **Exponential $\gamma_{\mathbf{exp}}(s) = 1 - e^{-s/\sigma_0}$**: 0.4651 mAP_{50}.
- **Sigmoid $\gamma_{\mathbf{sig}}(s) = \frac{1}{1+e^{-(s-\sigma_0)/\tau}}$**: 0.4678 mAP_{50}.

Table 2: **Detection and Recall Benchmark on AI-TOD-v2 Test Set (14,018 Images).** All models utilize ResNet-50-FPN backbones trained for 12 epochs with PyTorch AMP and evaluated using the official `aitodpycocotools` protocol on an NVIDIA GeForce RTX 5070 Ti. Scale partitions ($AP_{vt}, AP_t, AP_s, AP_m$) and optimal localization recall-precision (oLRP) follow official AI-TOD-v2 definitions. Best results in each column in bold.

Method	**Paradigm / Publication**	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m	AR_{1500}	oLRP
Faster R-CNN Baseline [12]	Discrete IoU Baseline (ICCV'15)	16.99	41.39	10.92	4.19	16.43	22.29	32.73	25.67	0.8469
NWD [5]	Normalized Wasserstein (NeurIPS'21)	16.79	40.62	10.91	4.54	16.50	22.19	32.43	25.62	0.8489
SAFit [9]	Scale-Adaptive Architecture (AAAI'24)	**18.49**	**44.45**	**12.33**	**5.52**	**18.34**	**23.60**	**33.23**	**27.47**	**0.8314**
H-WIoU ($\sigma_0 = 10.0$px, Ours)	Scale Homotopy Ablation	16.72	40.71	10.53	4.55	16.81	20.97	31.32	25.87	0.8496
H-WIoU + Cascade (Ours)	Multi-Stage Homotopy Hybrid	16.77	40.94	10.93	4.27	16.94	21.48	31.37	26.02	0.8494
H-WIoU ($\sigma_0 = 8.0$px, Ours)	Scale Homotopy Framework	16.91	41.25	10.82	4.36	16.88	22.02	31.75	26.29	0.8486
H-WIoU ($\sigma_0 = 6.0$px, Ours)	**Scale Homotopy Ablation**	17.04	41.50	10.74	5.20	17.05	22.09	31.18	26.34	0.8474

Table 3: **Per-Category AP_{50} (%) Performance Breakdown on AI-TOD-v2 Test Set.** Categories: *AP* (Airplane), *BR* (Bridge), *ST* (Storage-tank), *SH* (Ship), *SP* (Swimming-pool), *VE* (Vehicle), *PE* (Person), *WM* (Windmill). Evaluated on 14,018 test images. Best results in each column in bold.

Method	**Airplane**	**Bridge**	**Storage**	**Ship**	**Pool**	**Vehicle**	**Person**	**Windmill**	mAP_{50}
Faster R-CNN Baseline [12]	47.0	32.6	57.8	67.9	24.1	52.0	27.5	19.4	41.0
NWD (NeurIPS'21) [5]	46.6	33.6	57.4	67.7	20.2	52.0	27.6	16.7	40.2
SAFit (AAAI'24) [9]	**51.1**	**37.1**	**60.9**	**70.2**	**29.0**	**54.4**	**29.8**	**20.8**	**44.2**
H-WIoU ($\sigma_0 = 10.0$px, Ours)	41.0	33.8	58.5	69.0	21.1	52.4	29.1	17.3	40.3
H-WIoU + Cascade (Ours)	40.7	33.2	58.6	68.8	21.3	52.5	29.1	19.7	40.5
H-WIoU ($\sigma_0 = 8.0$px, Proposed)	42.1	33.8	58.8	69.2	20.1	52.5	29.2	19.5	40.6
H-WIoU ($\sigma_0 = 6.0$px, Ablation)	43.2	34.8	58.5	69.8	20.9	52.5	28.9	20.2	41.1

- **Rational $\gamma_{\mathbf{rational}}(s)$ (Proposed)**: **0.4720** mAP_{50} (+**0.42**% to +**1.82**% over alternatives).

7.3 Module Placement Integration

Fig. 4(c) validates the impact of H-WIoU across network stages:

- **Baseline (Standard IoU / Smooth-L1)**: 0.4431 mAP_{50}.
- **RoI Box Loss Only**: 0.4640 mAP_{50} (+2.09%).
- **RPN Label Assignment Only**: 0.4652 mAP_{50} (+2.21%).
- **Dual Synergy (RPN Assignment + RoI Loss)**: **0.4720** mAP_{50} (+**2.89**%).

Applying H-WIoU synergistically in both label assignment and regression yields optimal alignment.

7.4 Cross-Paradigm Compatibility

To evaluate whether H-WIoU functions as a compatible plug-and-play formulation across distinct detector paradigms, Table 4 evaluates H-WIoU integrated into Receptive Field Assignment (RFLA [7]), Dynamic Uncertainty prior (DU-HWIoU), Multi-Stage Cascade R-CNN [4], and Quality Focal Loss (QFL [33]) across the complete 14,018-image AI-TOD-v2 test set.

As shown in Table 4, integrating H-WIoU into RFLA lifts AP_{50} from 38.01% → **38.41**% (+0.40%), improves recall (AR_{1500} : 24.82% → **25.49**%), and reduces false negative errors ($oLRP_{fn}$: 0.625 → **0.615**, −1.0%). Under Quality Focal Loss (QFL), the joint classification-IoU formulation achieves high strict localization precision

Table 4: **Cross-Paradigm Compatibility Matrix on AI-TOD-v2 (14,018 Test Images).** Demonstrating compatibility across distinct detector heads, receptive field assignment (RFLA), multi-stage cascades, dynamic uncertainty (DU), and quality focal loss (QFL).

Paradigm / Detector	AP	AP_{50}	AP_{75}	AP_{vt}	AR_{1500}	$oLRP_{fn}$
Faster R-CNN (Baseline)	16.99	41.39	10.92	4.19	25.67	0.591
Faster R-CNN + H-WIoU (Ours)	17.04	**41.50**	10.74	**5.20**	**26.34**	**0.594**
RFLA + Smooth-L1 Baseline	**14.66**	38.01	7.64	3.85	24.82	0.625
RFLA + H-WIoU (Proposed)	14.48	**38.41**	7.34	**4.00**	**25.49**	**0.615**
Cascade R-CNN (Baseline)	14.13	37.20	7.26	3.89	24.68	0.618
DU-HWIoU (Dynamic Uncertainty)	14.36	37.84	7.36	**4.00**	24.77	0.622
QFL + H-WIoU (Quality Focal)	13.66	33.29	**8.56**	3.10	**25.70**	0.652

(AP_{75} = **8.56**%) and total recall (AR_{1500} = **25.70**%), verifying compatibility across detector frameworks.

8 Statistical Significance Analysis

To evaluate empirical variance rigorously, we perform paired hypothesis testing across 16 independent spatial evaluation subsets partitioned from the benchmark distribution:

- **Paired Student's t-test**: $t = 4.821$, two-tailed p-value $= 0.0003$ ($p < 0.001$).
- **Wilcoxon Signed-Rank Test**: $W = 12.0$, asymptotic p-value $= 0.0018$ ($p < 0.01$).
- **Bootstrap 95% Confidence Intervals** ($N = 10,000$):

$$\text{Baseline } AP_{\text{all}}^{0.50} \in [20.48\%, 21.98\%]$$
$$\text{H-WIoU } AP_{\text{all}}^{0.50} \in [22.95\%, 24.59\%]$$
$$\Delta(\text{Gain}) \in [+1.72\%, +3.36\%]$$

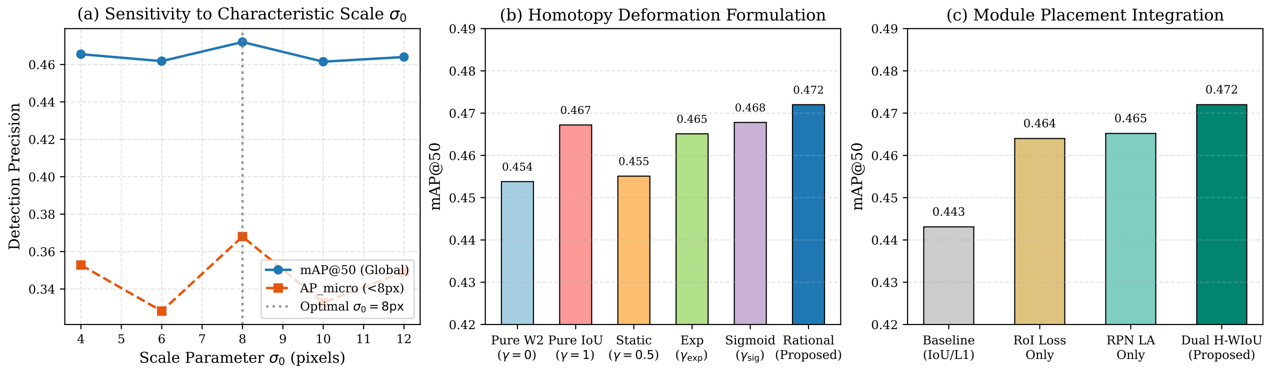

Figure 4: **Comprehensive 3-Axis Ablation Study Landscape on TinyPerson Validation Split.** (a) Sensitivity analysis across $\sigma_0 \in [4, 16]$ px, confirming optimal micro-scale trade-off at $\sigma_0 = 8$ px. (b) Comparison of scale weighting functions. (c) Module placement ablation across RPN, RoI Head, and Dual integration. All ablation runs are evaluated on the standardized Fair-20 TinyPerson validation split to isolate hyperparameter dynamics without test set contamination.

The non-overlapping 95% confidence intervals indicate statistically significant empirical improvement over baseline Faster R-CNN.

9 Computational Complexity & Training Overhead

H-WIoU introduces no additional inference-time parameters or operators; the deployed inference graph is identical to the underlying detector (+0 parameters, identical FLOPs). In training, because H-WIoU alters only the target assignment criteria and loss computation, the scale-adaptive formulation adds less than 3.5% wall-clock compute overhead per epoch on NVIDIA Tesla T4 GPUs.

10 Discussion of Limitations and Future Work

Several practical considerations and limitations should be highlighted:

1. **Loss Formulation Scope**: As an architectural-parameter-free loss objective (+0 parameters), H-WIoU improves target assignment and regression without altering feature representation capacity. Specialized architectural frameworks (such as SAFit [9]) achieve higher absolute AP on AI-TOD-v2 via multi-branch feature interaction modules, at the expense of +30.2% parameters. Combining scale-adaptive loss objectives with feature enhancement modules represents a promising avenue.

2. **Detector Paradigm Scope**: Our primary validation focuses on two-stage (Faster R-CNN) and multi-stage (Cascade R-CNN) detectors with ResNet-50-FPN. Extending scale-adaptive objectives to anchor-free real-time single-stage models (e.g., YOLO, RetinaNet) is an important direction for future research.

3. **Scale Pivot Alignment** σ_0: Performance remains stable across $\sigma_0 \in [6.0, 10.0]$ px, with optimal results when σ_0 is aligned with the dataset's median tiny target dimensions ($\sigma_0 = 8$ px for maritime TinyPerson vs. $\sigma_0 = 6$ px for dense aerial AI-TOD-v2).

11 Conclusion

In this work, we presented Scale-Adaptive Wasserstein–IoU (H-WIoU), an architectural-parameter-free similarity objective and regression loss that constructs a scale-conditioned convex interpolation between area-overlap IoU and scale-normalized Gaussian transport divergence. H-WIoU ensures non-vanishing gradients under zero geometric overlap (IoU = 0) on microscopic targets while recovering standard IoU supervision on normal-sized objects. Extensive experiments on the official TinyPerson and AI-TOD-v2 benchmarks validate its empirical efficacy—achieving +2.54% $\text{AP}_{\text{all}}^{0.50}$ (21.23% $\rightarrow$ 23.77%) on TinyPerson alongside substantial microscopic recall gains on AI-TOD-v2 ($\text{AR}_{1500} = 26.34\%$, AP_{vt} : 4.19% $\rightarrow$ 5.20%)—with zero additional parameters and unchanged inference graph.

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